Problem-Solution Fit

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Problem–Solution Fit Canvas

This canvas explains how the **Emotion Detection and Learning Support Engine (EDLSE)** addresses the emotional and learning challenges faced by students by identifying their problems, understanding the root causes, and providing an AI-powered solution that detects emotions and delivers personalized learning support.

Section	Details
1. Customer Segment (CS)	Students, Teachers, Educational Institutions, Online Learning Platforms, Corporate Training Organizations.
2. Problems / Pains (PR)	Students often experience stress, confusion, boredom, or frustration during learning. Traditional learning platforms cannot recognize learners' emotional states, resulting in reduced engagement and poor academic performance.
3. Triggers to Act (TR)	Increasing demand for personalized learning, emotional well-being monitoring, AI-assisted education, and adaptive learning systems.
4. Emotions (EM)	**Before:** Stress, anxiety, confusion, frustration, lack of motivation. **After:** Confidence, engagement, motivation, improved understanding, positive learning experience.
5. Available Solutions (AS)	Traditional Learning Management Systems (LMS), online quizzes, video lectures, counseling sessions, and manual teacher feedback. These solutions offer limited or no real-time emotion detection and personalized support.
6. Customer Limitations (CL)	Limited internet connectivity, low-end devices, privacy concerns, lack of emotional awareness tools, and limited access to personalized learning guidance.

Section	Details
7. Behavior (BE)	Students may lose concentration, skip difficult topics, become demotivated, or disengage from learning when experiencing negative emotions.
8. Channels of Behavior (CH)	**Online:** Web application, e-learning platforms, cloud-based systems, virtual classrooms. **Offline:** Smart classrooms, computer laboratories, educational workshops, and institutional deployments.
9. Problem Root Cause (RC)	Existing learning systems primarily measure academic performance and ignore learners' emotional states, making it difficult to provide timely support and personalized interventions.
10. Your Solution (SL)	An AI-powered application that captures facial expressions, voice, or text, detects the learner's emotional state using deep learning models, provides personalized learning recommendations, tracks emotional and academic progress, and generates detailed reports with actionable insights for students and educators.